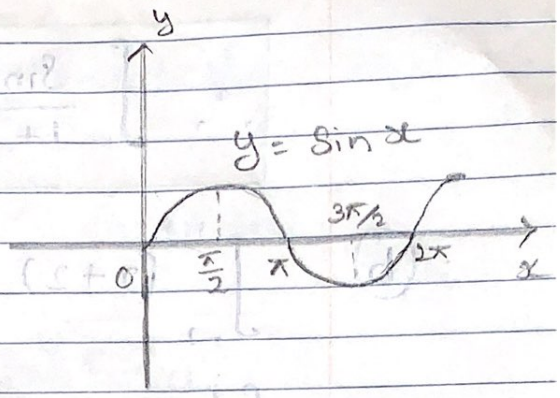
Math 401

(7) $\int_0^{\frac{3\pi}{2}} |\sin x|\, dx$



$$\int_0^{\frac{3\pi}{2}} |\sin x|\, dx$$

$$= \int_0^{\pi} \sin x\, dx + \int_{\pi}^{\frac{3\pi}{2}} -\sin x\, dx$$

$$= -\cos x \Big|_0^{\pi} + \cos x \Big|_{\pi}^{3\pi/2}$$

$$= -\cos\pi + \cos 0 + \cos 3\pi/2 - \cos\pi$$

$$= 1 + 1 + 0 + 1 = 3$$

$$\therefore \int_0^{\frac{3\pi}{2}} |\sin x|\, dx = 3$$

(8) (a) $\int \frac{\sin x}{1+\cos^2 x}\, dx$

Let $u = \cos x$, $du = -\sin x\, dx$

$$\int \frac{\sin x}{1+\cos x}\, dx = \int \frac{-du}{1+u^2}$$

$$= -\tan^{-1} u + C = -\tan^{-1}(\cos x) + C$$

$$\therefore \int \frac{\sin x}{1+\cos^2 x}\,dx = -\tan^{-1}(\cos x) + C$$

(b) $\int_{-1}^{1} (x+2)(\sqrt{1-x^2})\,dx$

$$\int_{-1}^{1} (x+2)(\sqrt{1-x^2})\,dx = \int_{-1}^{1} x\sqrt{1-x^2}\,dx + \int_{-1}^{1} 2\sqrt{1-x^2}\,dx$$

$$\Rightarrow \int_{-1}^{1} x\sqrt{1-x^2}\,dx + 2\int_{-1}^{1} \sqrt{1-x^2}\,dx$$

$$\int_{-1}^{1} x\sqrt{1-x^2}\,dx =$$

Let $u = 1-x^2$, $du = -2x\,dx$

$$= \int_{-1}^{1} \frac{-\sqrt{u}}{2}\,du = \frac{1}{2}\int_{-1}^{1} \sqrt{u}\,du$$

$$= \left(\frac{1}{2}\right)\left(\frac{2}{3}\right)\left(-u^{3/2}\right)\Big|_{-1}^{1} = \left(-\frac{1}{3}\right)(1) + \left(\frac{1}{3}\right)(1)$$

$$= 0$$

$$\therefore \int_{-1}^{1} x\sqrt{1-x^2}\,dx = 0$$

$$\int_{-1}^{1} (x+2)(\sqrt{1-x^2})\,dx = \int_{-1}^{1} 2\sqrt{1-x^2}\,dx$$

$$\int_{-1}^{1} 2\sqrt{1-x^2}\,dx = 2\int_{-1}^{1} \sqrt{1-x^2}\,dx$$

Let $x = \sin u$, $dx = \cos u\,du$

$$= 2\int_{-1}^{1} (\cos u)\left(\sqrt{1-\sin^2 u}\right) du$$

$$= 2\int_{-1}^{1} (\cos u)\left(\sqrt{\cos^2 u}\right) du$$

$$= 2\int_{-1}^{1} (\cos u)(\cos u)\,du = 2\int_{-1}^{1} \cos^2 u\,du$$

$$= 2\int_{-1}^{1} \frac{\cos(2u)+1}{2}\,du$$

$$= 2\int_{-1}^{1} \left(\frac{1}{2}\right)\left[\cos(2u)+1\right] du$$

$$= 2\int_{-1}^{1} \frac{1}{2}\cos(2u)\,du + 2\int_{-1}^{1} \left(\frac{1}{2}\right) 1\,du$$

$$= \int_{-1}^{1} \cos(2u)\,du + \int_{-1}^{1} 1\,du$$

Let $v = 2u$ and $du = \frac{1}{2}dv$ for equation ①

$$= \int_{-1}^{1} \left(\frac{1}{2}\right)\cos v\,dv + \int_{-1}^{1} 1\,dv$$

$$= \frac{1}{2}\int_{-1}^{1} \cos v \, dv + \int_{-1}^{1} 1 \, du$$

$$= \left(\frac{1}{2}\right) \sin v \Big|_{-1}^{1} + u \Big|_{-1}^{1}$$

$\boxed{v = 2u}$

$$\Rightarrow = \frac{1}{2} \sin 2u \Big|_{-1}^{1} + u \Big|_{-1}^{1}$$

$\boxed{u = \arcsin x}$

$$= \frac{1}{2} \sin(2 \arcsin x) \Big|_{-1}^{1} + \arcsin(x) \Big|_{-1}^{1}$$

$$= \frac{2x\sqrt{1-x^2}}{2} \Big|_{-1}^{1} + \arcsin(x) \Big|_{-1}^{1}$$

$$= (1)\left(\sqrt{1-(1)^2}\right) - (-1)\left(\sqrt{1-(-1)^2}\right) +$$

$$+ \arcsin(1) - \arcsin(-1)$$

$$= 0 - 0 + \frac{\pi}{2} + \frac{\pi}{2} = \frac{\pi}{2} + \frac{\pi}{2}$$

$$= \pi$$

$$\therefore \int_{-1}^{1} (x+2)\left(\sqrt{1-x^2}\right) dx = \pi$$